ROOT CLASSES - MATHEMATICS

Class 10th NCERT Study Material & Formulas

Chapter 12: Surface Areas and Volumes

Key Concepts & Important Formulas:

* Cylinder: CSA = 2*pi*r*h, TSA = 2*pi*r(r + h), Volume = pi*r^2*h.
* Cone: Slant height l = sqrt(r^2 + h^2), CSA = pi*r*l, TSA = pi*r(r + l), Volume = (1/3)*pi*r^2*h.
* Sphere: Surface Area = 4*pi*r^2, Volume = (4/3)*pi*r^3.
* Hemisphere: CSA = 2*pi*r^2, TSA = 3*pi*r^2, Volume = (2/3)*pi*r^3.
* Combination of Solids: Total volume is sum of volumes; total surface area is exposed surface area.

Solved Exam Examples & Practice:

Example: A solid toy is in form of hemisphere surmounted by cone of same radius 3.5 cm. Total height is 15.5 cm. Find volume of toy.